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Generative AI and the Future of Work

In May 2023, 11,500 screenwriters in the Writer’s Guild of America (WGA) union went on strike, halting work on television and film productions. The document outlining the union’s position on key issues read: “Regulate use of artificial intelligence on MBA [Minimum Basic Agreement]-covered projects: AI can’t write or rewrite literary material; can’t be used as source material; and MBA-covered material can’t be used to train AI.”¹ Though many issues were at stake, the terms governing the use of generative AI in the creative process turned out to be one of the leading issues in the negotiations.

As the strike went on through the summer of 2023, writers everywhere acquainted themselves with the new technology and stared down a version of the future in which they were increasingly replaceable. In October, as part of the deal that eventually ended the strike, WGA negotiators and the studios agreed on four new terms governing the use of artificial intelligence on union projects, including provisions that “AI-generated material will not be considered source material,” and “AI-generated material can’t be used to undermine a writer’s credit or separated rights.”²

By tying the writer’s credit system firmly to the human writer, allowing the writer and employer the flexibility to decide the role generative tools will play in their own creative processes, and reserving the right to decide its position on the question of using copyrighted materials in training at a later date, the first-ever explicit agreement between management and unionized workers over the terms of generative AI in the workplace satisfied the union’s most important concerns and helped get the striking writers back on the job.

Screenwriters and headline writers alike considered the deal a win. Reflecting the concerns of their readerships, coverage in major publications focused on the implications of the deal for the future of work in other creative and professional fields. In the days following the resolution, *The Los Angeles Times* ran a column headlined: “The writers’ strike was the first workplace battle between humans and AI. The humans won.”³ *Wired* headlined its coverage: “Hollywood writers reached an AI deal that will rewrite history,”⁴ and *The New York Times* ran a guest essay with the headline, “Hollywood’s deal with screenwriters just rewrote the rules around A.I.”⁵

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However, there still remained legal issues around the use of generative AI in the workplace. In July 2023, a group of authors filed a class-action lawsuit against OpenAI, accusing it of illegally using their copyrighted works to train its language models. The authors argued that these models were producing infringing, derivative works, and that OpenAI had violated intellectual property laws by training its models on unlicensed data.⁶ While most copyright claims were dismissed in court, the case highlighted unresolved legal questions, leaving creatives uncertain about their ability to protect their work in the future.⁷ Beyond copyright concerns, broader regulatory challenges around issues like data privacy, bias, and AI accountability continue to shape the legal landscape, with governments around the world working to establish frameworks for responsible AI development and use.⁸

In one sense, the WGA deal seemed to spell a new era for human-AI labor relations. In another sense, the drama of humans and labor-saving AI learning to co-exist in society, the workplace, and the economy at large was just getting started.

Generative AI

On November 30, 2022, OpenAI released ChatGPT, a text-generating chatbot based on a large language model (LLM) it had developed. Large language models are AI systems trained on vast amounts of text data to understand and generate human-like language. Though increasingly-capable LLMs had been used in AI labs for years, ChatGPT was among the first generative tools to capture mainstream attention, thanks in part to its intuitive chatbot-style and availability to the general public.⁹ The chatbot appeared to effectively parse queries and formulate human-sounding outputs on almost any imaginable topic in a range of formats, from explanatory prose to creative limerick poetry to complex legal arguments.¹⁰ ChatGPT reached over one million users in its first week,¹¹ and over 100 million users in two months.¹²

As the public acquainted itself with the new capabilities of generative AI, various narratives about the technology's impact swirled in the media and popular culture. Generative AI's potential to reshape fundamental aspects of work and society quickly became a prominent topic of discussion, sparking widespread debates about its impact on the future. Responses to the technology reflected a broad spectrum of concerns and speculations about how it might alter industries, human creativity, and the structure of labor.

For perhaps the first time, white-collar professionals were faced with the prospect that automation could come for a percentage of their jobs. If AI could effectively absorb data, analyze it, and produce compelling written output, what would become of the text-producing white-collar creative class of writers, professors, strategists, marketers, consultants, researchers, journalists, securities analysts, managers, and lawyers, not to mention the code-producing class of programmers and software engineers? Though downstream from text in terms of immediate impact, image-, sound-, and video-producing professionals experimenting at the frontiers of audio and image-generating AIs quickly learned that their own fields would be similarly exposed to the generative AI revolution in the coming years.

How it Began - Natural Language Processing

Modern language models evolved from Natural Language Processing (NLP), a field that dates back to the early days of computer science. Initial efforts to teach machines to understand human language began in the 1950s and 60s. One early example was the Georgetown-IBM experiment (1954), which demonstrated machine translation of Russian sentences into English using a fixed glossary and translation rules.¹³ Around the same time, the Dartmouth Summer Research Project on Artificial

Intelligence (1956) gathered leading thinkers in the nascent fields of artificial intelligence, information theory, and cybernetics, clarifying the boundaries, key questions, and subfields of AI. The conference, which was attended by AI and computing pioneers Marvin Minsky, Claude Shannon, and Nathaniel Rochester, is widely considered the founding event for the field of artificial intelligence.¹⁴

MIT Computer Scientist Joseph Weizenbaum's 1966 ELIZA program was another early milestone in the use of NLP and conversational interfaces. Programmed to reply to a user's text inputs in the conversational manner of a Rogerian therapist ("That's interesting, tell me more"), the replies the program generated were based on a set of simple pattern-matching rules. Though rudimentary by today's standards, ELIZA is important as one of the first-known examples of the genre of interactive software we now call a chatbot.¹⁵

These early NLP efforts were rule-based, and therefore required manual scripting of linguistic rules, an approach that was limited by the complexity and nuance of human language. As computational capacities grew, so did interest in more creative and dynamic approaches. In the 1990s and early 2000s, statistical methods became the cutting edge of NLP research. IBM's Candide (1994), one of the first statistical machine translation systems, was an explicit departure from the rule-based approach, and its success helped reorient the field toward the use of big data and statistical methods.¹⁶

Statistical models like Candide learned from vast amounts of text data and calculated translation possibilities according to their probability, making them more robust and adaptable than the rule-based systems of the earlier era. The first version of Google Translate, launched in 2006, advanced and extended the use of statistical machine translation, relying on vast amounts of bilingual text data (sourced in part from the EU, Canada, and other government bodies where legally required translation and publication of laws and proceedings had produced vast corpora of multilingual matched texts) to derive statistical probabilities for translation.¹⁷

Neural Networks

Neural networks are a type of artificial intelligence modeled after the human brain's structure, designed to recognize patterns and make decisions. They consist of layers of nodes – individual units that process inputs, apply weights, and generate outputs. These networks "learn" by adjusting the weights between nodes based on errors, refining their ability to classify and predict outcomes. Neural networks were first introduced in the 1950s, but struggled to gain traction due to limited computational power and inadequate data.

Statistical methods remained the state of the art in AI until the 2010s, when researchers at Google Brain revitalized the neural network paradigm through deep learning – an approach that uses multiple layers of neurons to learn and classify patterns from vast datasets across various levels of abstraction.¹⁸ Google Brain's defining approach was to model artificial intelligence after the human brain. Their work helped popularize the concept of the node, modeled after the neuron, as the fundamental unit of an AI system.¹⁹ Just as biological neurons receive signals, process them, and transmit outputs to other neurons, nodes in an artificial neural network receive an input, apply a series of weights, activate (or not), and produce a mathematical output that is passed along to another node where the process is repeated. Nodes are nested in layers, the bulk of which (such as those performing the actual classification, clustering, or learning) are hidden from the user.²⁰

Geoffrey Hinton, a professor at the University of Toronto who has been called "the Godfather of AI"²¹, did pioneering work on backpropagation algorithms for neural networks in the 1990s and early 2000s, before they became widespread in AI circles. Backpropagation is a training technique for neural networks that involves taking the error rate from the previous training cycle and feeding it back

through the network to adjust the weights on the next iteration.²² Hinton began working at Google in 2013 after his company DNNResearch was acquired by the search giant.²³ Hinton's contributions helped set the stage for the great leap forward in AI progress in the second half of the 2010s.²⁴

Transformer Architecture

The next major wave of AI was driven by the introduction of the transformer architecture by researchers at Google Brain in 2017.²⁵ The transformer architecture employed a novel self-attention mechanism that allowed the model to process inputs in a holistic, parallelizable way.

To understand how a transformer works, imagine a CEO calling their direct reports into a room and, following an agenda, gets updates from each department head in order. The CEO decides which updates are most important to focus on. They may stray from the agenda to ask follow-up questions, but each digression takes additional time. Now, imagine the CEO has the upgraded capability to process auditory information in parallel. A 30-minute meeting is cut to five minutes. The CEO hears everything at once, makes the right connections between the individual bits of information, and ends the meeting by giving each department head instructions informed by the relevant information in each update.

The upgraded CEO represents the advance between the processing capacity of pre- and post-transformer neural networks. Behind the scenes, the transformer processes the inputs by converting them into a matrix of weighted values that capture the relationships between components. It then uses these weights to determine which elements to focus on next. Thus, training is faster than with linear methods, and the resulting models are better at abstracting and generalized reasoning across contexts in a human-like way.²⁶

By the end of the 2010s, the self-attentive transformer architecture had become the bedrock for a new generation of large-scale language models. Its success in training larger and larger models led to a flurry of attention in the corporate world. Tech companies began investing heavily in AI and LLM R&D, hiring teams of experts to explore how to optimize and scale up transformer-based models. In 2019, Microsoft invested \$1 billion in the newly restructured capped-profit arm of OpenAI.²⁷

From Transformers to ChatGPT

Building on the foundations laid by the transformer architecture, OpenAI introduced a series of models under the GPT (Generative Pre-trained Transformer) banner. GPT models are trained on vast amounts of data to generate human-like text by predicting the next word or phrase in a sequence. OpenAI's first GPT iteration showed researchers the technology's potential. But it was the release of GPT-2, which showcased an impressive ability to generate coherent and contextually relevant text over long passages, that took many by surprise. At the same time, the unexpected power of the model raised concerns about the potential for misuse, leading OpenAI to initially withhold it from the public.²⁸

By the time GPT-3 arrived, it was clear that the NLP landscape had changed. GPT-3 boasted 175 billion parameters – weights that the model adjusts to predict the next word – compared to GPT-2's 1.5 billion. This massive increase allowed GPT-3 to understand and generate text with much greater accuracy. More notably, GPT-3 could perform entirely new tasks upon the first request, without any task-specific fine-tuning, by leveraging its vast pre-existing knowledge and ability to generalize. This capability, known as “zero-shot” learning, represented a major leap forward in AI's flexibility and adaptability.^{29, 30} As word of GPT-3's emergent abilities made its way to the executive suite, businesses began seeing the potential of generative AI for a range of applications: customer service automation, content creation, coding assistance, and more.

ChatGPT, a chatbot built on OpenAI's GPT-3.5 model, was released in November 2022 and immediately captured the public's attention. ChatGPT was designed with a user-friendly conversational interface, making advanced AI capabilities easily accessible to non-technical users. This accessibility, combined with its impressive ability to generate natural-sounding dialogue, led to ChatGPT's rapid adoption. By January 2023, it had become the fastest-growing consumer app of all time.³¹ In February 2023, OpenAI began offering ChatGPT Plus as a subscription service with additional features for \$20 a month.³² In March 2023, OpenAI released GPT-4 to ChatGPT Plus and API subscribers.³³ Experts estimated that the model underlying GPT-4 had over 1.7 trillion parameters—more than 10 times the size of GPT-3.³⁴ Consequently, GPT-4 demonstrated substantial performance gains, notably improving its ability to generate coherent and contextually relevant responses with up to 40% better accuracy on complex benchmarks compared to GPT-3.³⁵

ChatGPT's high-profile success in the consumer market spurred further innovations and a growing interest in pushing the technological and commercial boundaries of the field. Within the next year, Microsoft announced plans to integrate generative "copilots" into its Microsoft 365 suite of productivity software, Adobe embedded a generative model and AI tools into its various design apps, and Google began work integrating its Gemini chatbot into Chrome and Android.³⁶

Beyond Static Data: Retrieval-Augmented Generation and Fine-Tuning

Despite significant advancements in model accuracy, AI hallucinations remained a major concern. Hallucinations—instances where models confidently produced false or misleading information—occurred because GPT models, such as GPT-3 and GPT-4, were designed to predict the next word or sentence based on patterns in their training data, without any mechanism to verify the accuracy of their outputs. Further, these models were reliant on static pre-trained datasets and could not access real-time information or external knowledge. As a result, when prompted with queries outside the scope of their training, these models often fabricated plausible but inaccurate responses. This limitation raised concerns over the reliability of LLMs in critical or specialized applications.

Retrieval-Augmented Generation (RAG) was developed as an approach to addressing these challenges. RAG combines LLMs with external knowledge sources through a vector database—a system that stores unstructured data like research papers or news articles as numerical vectors. Unlike traditional LLMs that relied solely on static, pre-trained datasets, RAG-based models can access real-time, up-to-date information from external sources, including the internet or enterprise databases, at the time of the query.^{37, 38} This ability to retrieve new information made RAG-based systems far more dynamic and reduced the risk of generating outdated or incorrect responses. Businesses, especially those in customer service and technical support, leveraged RAG to provide accurate, context-aware, and personalized responses that reflect the latest available data.

Another method for improving LLM accuracy is fine-tuning, which involves updating the model's parameters with domain-specific data to improve its performance in specialized areas. For example, BloombergGPT, which was trained on general data and fine-tuned for the financial sector, outperformed other models on finance-related tasks at the time of its release. However, while fine-tuning provides high accuracy in specific domains, it relies on static datasets and therefore lacks the flexibility of RAG. The tradeoff between the approaches lies in fixed versus variable costs: fine-tuning requires a larger initial investment, while RAG incurs ongoing computational costs as it retrieves information on demand.

Next Steps in AI Development

By late 2024, several new trends in AI development were beginning to shape the future of the technology. AI agents—autonomous systems that could actively retrieve information, schedule, and trigger workflows based on inputs or goals—were pushing AI beyond text generation and into real-world applications. Similarly, multi-modal models, capable of processing and generating data across text, images, and audio, were enhancing fields such as medical diagnostics and creative industries by allowing AI to work seamlessly across different data types.

One of the most significant advancements was Chain of Thought (CoT) reasoning, a technique that allowed models to break down complex problems into smaller steps.³⁹ OpenAI's 4-o1 release, which incorporated CoT prompting, reinforcement learning, and error-checking, exemplified that increasing computational power in reasoning tasks led to step-change improvements in performance.⁴⁰ These updated models were able to reason more effectively than their predecessors, demonstrating improved accuracy in complex decision-making and problem-solving.⁴¹ These developments also contributed to increased transparency in AI models, reducing the “black-box” nature of earlier versions. CoT allowed users to follow the reasoning process step by step, while human-led reinforcement learning helped develop reasoning that more closely mirrored human thought processes.⁴²

Another key shift by late 2024 was the move away from development strategies that were primarily focused on increasing the number of model parameters. While increasing parameter counts had significant effects on accuracy in early upgrades of large language models, the marginal returns of adding parameters began to diminish past a certain level.⁴³ Techniques like CoT and RAG offered more efficient improvements in performance without the ballooning costs and memory usage associated with building continuously larger models. These innovations set the stage for the next generation of AI, especially in fields requiring deep, task-specific reasoning.

Ethical and Safety Concerns in AI Development

As generative AI technology advanced in popularity and capability, concerns over its ethical and societal implications became increasingly prominent.⁴⁴ Researchers shifted their focus from merely scaling models to ensuring that they were aligned, interpretable, and safe for public use. AI developers—including OpenAI, Microsoft, Anthropic, and Google—implemented various safety measures and guardrails in their publicly available models and chatbots, aiming to reduce biases and prevent harmful outputs.⁴⁵

Long-term safety was also a concern. In March 2023, an open letter signed by Elon Musk, Steve Wozniak, and over 1,000 other technology leaders called for a temporary pause in AI development, citing “profound risks to society,” such as the potential for job displacement, misuse of AI in disinformation, or applications in military settings.⁴⁶ In May 2023, Geoffrey Hinton resigned from Google to raise awareness about AI risks.⁴⁷ Hinton became the first signatory of a statement by the Center for AI Safety, which read, “Mitigating the risk of extinction from AI should be a global priority alongside other societal-scale risks such as pandemics and nuclear war.” The statement was co-signed by a group of industry leaders, including Sam Altman of OpenAI, Demis Hassabis of Google DeepMind, and Dario Amodei of Anthropic.⁴⁸ Further capturing these fears, a 2024 survey of over 2,700 prominent AI researchers found that 47% of respondents estimated at least a 10% chance of AI contributing to human extinction.⁴⁹ These concerns reflected broad fears that unchecked AI development could lead to destabilizing critical infrastructure or enabling large-scale autonomous weapons, which could, under worst-case scenarios, pose existential threats to humanity.

Impact on Labor and Employment

As generative AI advanced, its impact on labor and employment was poised to echo the double-edged sword of past technological revolutions—displacing some jobs, creating new ones, and reshaping others. During the mechanization of textile production in the Industrial Revolution, for example, many weavers and artisans were displaced, but new roles emerged in machine oversight and specialized tasks like sewing cuffs, collars, and buttons. Mechanization did not eliminate labor but redefined it, as workers transitioned from crafting complete items to becoming part of a larger industrial process.⁵⁰ However, in spite of the overall increase in productivity brought by the new systems, some historians assert that the monotonous, tedious, and often hazardous nature of the new jobs, combined with the unequal power dynamics between employers and employees, left many workers worse off under the new system.^{51, 52}

Similarly, rather than exclusively displacing human labor, generative AI was expected to drive a process of coevolution, where human roles adapted alongside AI, creating new forms of work that blended human oversight with AI-driven processes. While AI promised increased efficiency, it also raised concerns about job quality. Like early industrial jobs, some roles created by AI were predicted to lack the autonomy or fulfillment of earlier jobs. These historical trends therefore highlighted the importance of managing AI's integration thoughtfully to ensure that new opportunities optimized productivity gains without sacrificing worker well-being.

Commercial Implementation: Video Games and Generative AI

After ChatGPT's release to the general public in fall of 2022, companies across a variety of sectors raced to implement generative tools, revolutionizing internal business operations and consumer experiences alike. Despite being largely insulated from the impacts of automation in the past, creative fields now sought out opportunities to integrate generative technology. The video game industry, a \$217 billion global market as of 2022, quickly began finding applications for AI in game development.⁵³

The video game company Electronic Arts, for example, announced in March 2024 that it was incorporating generative tools in up to 60% of its development process, and expected a 30% increase in efficiency. Notably, the announcement came one week after the company laid off 670 employees – about 5% of their workforce.⁵⁴ By July, EA expressed that generative AI had been integral to creating *College Football 25*, allowing them to produce over sixteen thousand character models – all based on real players, fans, and coaches – within five months.⁵⁵ Other video game developers including Microsoft had similarly announced that generative AI would become essential to their game development processes, especially in localization, microtransactions, and game testing.

While many game developers promised that AI integration wouldn't lead to significant job displacement, an estimated 10,500 employees were laid off across the industry in 2023, and 11,000 more had been laid off by the summer of 2024. Some industry professionals reflected that while AI might not yet possess the creativity needed to replace truly great art, it was already capable of producing work that was “good enough” to meet commercial standards, and therefore already capable of replacing workers in certain roles.⁵⁶ In July 2024, over 160,000 video game voice and motion capture actors in the SAG-AFTRA union announced a strike, demanding “critical AI protections” for union members, of the sort that WGA screenwriters had won the summer before.⁵⁷

The Future of (Human) Jobs

The rollout of generative AI tools in the early 2020s created widespread anxiety about the trajectory of the technology and what its eventual capabilities might do – both to human labor and the social contract more broadly. The specter of artificial general intelligence (AGI) – non-human intelligence that could reason, learn, and act with autonomy across domains without human intervention – loomed large in the debates between “accelerationists” (advocates for pushing AI technology forward as quickly as possible) and “doomers” (advocates for safety and alignment). Among the immediate fears were unemployment and the potential destabilization of societies.

Many of these observers’ concerns about generative AI’s impact on labor and employment mirrored those raised in the Industrial Revolution: de-skilling of jobs, downward pressure on wages, increased surveillance, and diminished bargaining power of human workers in the face of competition from machines newly able to do the same type of work. Would machines take over at least some of the daily work tasks, both cognitive and physical, that some humans enjoy and find meaning in? If so, how would humans respond and adapt?

One possibility was that fewer humans would need to work. The World Economic Forum’s 2023 “Future of Jobs Report,” based on surveys of more than 800 companies, found that employers expected a net reduction of 14 million jobs, or 2% of current employment, by 2027, due in part to labor-saving technologies resulting from advances in AI and automation.⁵⁸ A year later, staff at the IMF speculated that up to 70% of all jobs in advanced economies such as the United States were exposed to some degree of AI automation, and up to 1.4 billion full time jobs could be seriously impacted globally.⁵⁹ Policies ranging from workforce re-skilling programs to universal basic income were proposed and debated as potential solutions to the societal and economic shifts that more widespread AI might entail.^{60, 61}

Forecasting how AI might impact the workplace is potentially fraught work. In the past, even reputable organizations such as McKinsey and Company have overestimated the workplace impact of new technologies by orders of magnitude.⁶² Though the precise impacts of generative AI on the future of work were uncertain, recent academic work offered valuable frameworks.

“Automation and New Tasks: How Technology Displaces and Reinstates Labor,”
Acemoglu and Restrepo (2019)

Before the surge of interest in generative AI following ChatGPT’s release, Acemoglu and Restrepo (2019) examined key labor market issues during periods of technological change. They present several key historical insights.

Historically, technology displaces and reinstates labor. “The history of technology is not only about the displacement of human labor by automation technologies... Instead, the displacement effect of automation has been counterbalanced by technologies that create new tasks in which labor has a comparative advantage. Such new tasks generate not only a positive productivity effect but also a reinstatement effect – they reinstate labor into a broader range of tasks and thus change the task content of production in favor of labor. The reinstatement effect is the polar opposite of the displacement effect and directly increases the labor share as well as labor demand.”⁶³

Effect of automation on labor demand = Productivity effect + Displacement effect. Automation enables capital to substitute for labor in certain tasks, increasing productivity by allowing cheaper substitutes to perform previously expensive tasks at a lower cost (i.e., machines or unskilled workers substituting for expensive skilled workers). The productivity effect makes the producer more efficient while shifting labor demand to new, non-automated tasks where labor demand is higher. The

displacement effect reduces the demand for one type of labor, usually expensive skilled labor, reducing labor's share of the value added (labor share of value is highest when skilled labor is both scarce and required – as with, say, the current job market for AI engineers). The net impact depends on whether the economic value of the productivity gained outweighs the economic value of the labor displaced. As the authors say: “Automation therefore increases the size of the pie, but labor gets a smaller slice. There is no guarantee that the productivity effect is greater than the displacement effect; some automation technologies can reduce labor demand even as they raise productivity.”⁶⁴

The net impact on labor demand depends on the broader labor market. “Different technologies are accompanied by productivity effects of varying magnitudes, and hence, each set of automation technologies may impact labor demand in different ways. Likewise, because the productivity gains of automation depend on the wage, the net impact of automation on labor demand will depend on the broader labor market context. When wages are high and labor is scarce, automation will generate a strong productivity effect and will tend to raise labor demand. When wages are low and labor is abundant, automation will bring only modest productivity benefits and can end up reducing labor demand.”⁶⁵

In settings where wages are already low, it can be hard to squeeze enough productivity out of the new technology for there to be any net productivity gain at all, let alone enough to spur an overall increase in labor demand from reallocating displaced workers into new tasks. Take the example of self-checkout machines at grocery stores. Wages for grocery store clerks are already low, so automating away the human jobs might result in limited economic productivity gains. Self-checkout machines would, in that case, reflect “so-so automation” – technologies that replace workers without offering any real productivity benefits. Self-checkout machines simply transform paid labor by a store employee into unpaid labor by the customer, who scans and bags their own groceries. The store may save a bit on labor costs if the machines can be installed and maintained for less than the clerk's salary, but the clerk's salary – transformed into a capital investment – disappears from the wider economy. Labor demand is reduced, net productivity is flat, and customers now bag their own groceries.⁶⁶

Recent waves of automation are not creating enough new tasks. Historically, automation's displacement of labor has been counterbalanced by the creation of new tasks where human labor has an advantage. Yet, according to the authors, much of the automation in recent years has created only minor productivity benefits while new task creation has slowed. The authors suggest that an increasing focus on maximizing profits for shareholders, combined with the rise of flexible employment arrangements and government policies that favor capital over labor, have allowed the new task creation that accompanies displacement to stall. A related possibility is that while some new tasks are being created at the leading edge of the economy, these jobs are fewer in number and less accessible to ordinary workers than in past eras because of skill and education gaps. Societies have continued to encourage the automation of expensive tasks using technology, but the replaced workers have not been reinstated in the kinds of new, non-automated tasks that have accompanied other periods of rapid technological advancement.⁶⁷

In a 2024 essay building off of his work with Restrepo, Acemoglu concludes that AI has a high potential for continuing the trend of “bad” automation that displaces workers without creating new tasks or meaningful productivity gains. Specifically, he argues that, while AI-driven automation may result in modest productivity gains (0.71% over the next 10 years), these gains are likely overstated because they come primarily from tasks that are easier to automate. Meanwhile, the creation of new tasks that could benefit workers has slowed, exacerbating inequality and concentrating economic gains among capital owners rather than workers.⁶⁸

“GPTs are GPTs,” Eloundou et al. (2023)

Building on prior work examining the impact of machine learning technologies on jobs, a paper by researchers at OpenAI and the University of Pennsylvania compared the particular tasks performed within occupations to the set of tasks that could be readily automated with LLMs. They then aggregated the occupation-level exposure to generative AI to estimate the possible impact of the diffusion of LLM technology on the U.S. labor force.⁶⁹

Half of all US jobs could be exposed. Across all jobs, about 15% of tasks were directly exposed to LLMs. The researchers found that about 80% of the workforce could have about 10% of their current job tasks exposed, while approximately 20% could have at least 50% of their tasks exposed. Those figures measured only the tasks directly exposed to automation by core LLM functionality. When software, tools, and interfaces built on top of LLMs (such as ChatGPT and coding assistants) were factored in, the percentage of U.S. job tasks exposed to generative AI increased to between 47 and 56%.⁷⁰

Higher paying jobs are more exposed. Researchers found that jobs with manual labor components would be less affected by LLMs, while jobs heavy in coding and writing tasks were relatively more exposed. By sector, jobs in manufacturing, agriculture, and mining had lower exposure, while jobs in information-processing industries were more exposed. By income distribution, jobs with higher wages showed higher exposure than lower-wage jobs (see **Exhibit 1**).⁷¹ They summarized:

If ‘GPTs [Generative Pre-trained Transformers] are GPTs [general-purpose technologies],’ the eventual trajectory of LLM development and application may be challenging for policymakers to predict and regulate. As with other general-purpose technologies, much of these algorithms’ potential will emerge across a broad range of economically valuable use cases, including the creation of new types of work.⁷²

“Generative AI at Work,” Brynjolfsson, Li, and Raymond (2023)

In a field study conducted at the customer contact center of a Fortune 500 company, researchers measured the impact of access to a generative AI tool on contact-center worker productivity. Customer service agents interacted with customers through a chat-based interface while the AI tool, in a side window, processed the interaction in real-time and suggested responses, essentially giving agents a highly-capable predictive autocomplete. Agents were free to use or ignore the AI assistant’s suggestions. Researchers gave one group of agents AI access and compared their productivity to a control group that had no tool available.⁷³

Access to generative AI tools increased productivity. Using resolutions per hour as the primary measure of productivity, researchers found that agents with access to the tool resolved 14% more cases per hour than agents without the tool (see **Exhibit 2**). Agents using the tool also exhibited lower average handle time, an increase in chats per hour, and higher customer satisfaction scores as compared with the control group. In subsequent analyses, access to AI assistance improved customer sentiment, reduced requests for managerial intervention, and improved employee retention.⁷⁴

Lower-skilled workers benefited more. The researchers used agents’ pre-intervention productivity metrics to compare pre-and post-intervention productivity at the individual and group levels. Using the pre-intervention productivity baselines, they were able to explore the impact of access to the generative tool across the distribution of agent productivity. Those in the lowest 20% of the pre-experiment productivity distribution saw a 35% increase in resolutions per hour, while workers in the highest skill quintile saw no significant change in resolutions per hour (see **Exhibit 3**).⁷⁵

“Navigating the Jagged Technological Frontier,” Dell’Acqua et al. (2023)

In a study conducted with Boston Consulting Group, a team of researchers examined the impact of access to generative AI on highly-skilled knowledge workers. A total of 758 consultants, all aware they were participating in a study, were divided between two experiments, one designed to test the impact of generative AI on tasks researchers considered “inside the frontier” of generative AI at the time, and a second experiment designed to test the impact of generative AI on tasks considered “outside the frontier” of generative AI. This was one of the first studies to examine the impact of generative AI on high-skill professionals engaged in primarily creative knowledge work.⁷⁶

Inside the frontier, professionals with generative AI were more productive. Consultants were pre-tested on a similar task to establish a baseline, then given tasks across four categories considered to be within the boundary of existing generative AI capability: Creativity (e.g., “Propose at least 10 ideas for a new shoe targeting an underserved market or sport.”); Analytical thinking (e.g., “Segment the footwear industry market based on users”); Writing proficiency (e.g., “Draft a press release marketing copy for your product”); and Persuasiveness (e.g., “Pen an inspirational memo to employees detailing why your product would outshine competitors.”) Consultants with access to generative AI finished 12% more tasks, completed those tasks 25% more quickly, and produced 40% higher quality results than those without AI access (see **Exhibit 4**).

Inside the frontier, lower-skilled professionals benefitted more. Though all consultants in the AI group saw an increase in performance, consultants who pre-tested in the bottom half of the pre-intervention skills distribution saw a greater increase (43%) than consultants who pre-tested in the top half (17%) (see **Exhibit 5**). One possible explanation is that generative AI is inherently limited to reproducing outputs that align with what its trainers have defined as “high-quality.” Since these benchmarks are based on the performance of top consultants, AI lacks the capacity to push the performance of already high-achieving individuals significantly further.⁷⁷

Outside the frontier, professionals with generative AI performed worse. Simulating case interviews used in recruiting, consultants were asked to make a product decision based on an analysis of qualitative and quantitative data. This task was presumed to be outside the frontier. The authors designed the task to have only one correct answer to simplify evaluation. In this experiment, the assistance of generative AI hindered, rather than helped performance. Consultants with no AI access got the right solution about 84% of the time, while consultants with AI access got the right answer only about 65% of the time. The authors speculate that the reduced performance of consultants with AI may have resulted from their trusting the AI assistant too much and “falling asleep at the wheel.”⁷⁸

The capabilities of generative AI are a jagged frontier. In practice, it is hard to know exactly where the capabilities of generative AI begin and end. The principal metaphor of the paper, “the jagged frontier,” is intended to illustrate this fact (see **Exhibit 6**). Some tasks that seemed like they should be inside the frontier were actually outside it. For example, many LLMs in 2023 struggled with simple arithmetic due to their reliance on pattern recognition rather than logic or calculation (later chatbots addressed this issue by integrating modules to handle mathematical reasoning). Meanwhile, some things that seemed like they should be outside the frontier could be done surprisingly well by generative AI—generating ideas for a new product name, or putting together an itinerary for an upcoming trip. These expectations shift as users gain experience. Ultimately, the authors remind, it is impossible to know where the frontier lies on any given task without experimentation.⁷⁹

Successful generative AI users were centaurs or cyborgs. Across both studies, the researchers observed two patterns of navigating the jagged frontier of generative AI: “centaurs” and “cyborgs.”

Centaurs, named for the half-human/half-horse creatures of Greek mythology, used a strategic division of labor between human and machine, allocating tasks to either themselves or to the generative AI on a task-by-task basis depending on their own strengths and the tasks they could confidently hand off to AI. For example, a consultant who was more confident in their analytical skill than in their writing might hand off most of the writing tasks to the AI, while leading the analysis themselves. Cyborgs, named for the integration of man and machine into one body, constantly experimented at the boundaries of the system's capabilities, intertwining generative AI into their workflows at every step. The study did not compare the effectiveness of the two approaches.⁸⁰

Social Implications and Policy Considerations

The rise of generative AI was expected to create societal-level shifts reminiscent of past technological revolutions. The Industrial Revolution, for example, precipitated major social shifts including urbanization, changes to family life, class divisions, and cultural upheavals as traditional communities and lifestyles were upended.⁸¹ The technological path of industrialization helped to create a new class of workers who lived in cities and had new sources of contact with books, ideas, and elites. Some believed the interplay of new economic conditions, physical conditions, and class consciousness helped give rise to new political movements and ideologies, from labor organizing and unions to Marxism and communism.

Similarly, in the late 20th century, technological improvements in telecommunications, logistics, and shipping paved the way for another macro trend: globalization. Some historians saw globalization, rooted in the economic logic of flexible labor markets, free trade, and comparative advantage, as responsible for hollowing out the industrial base in wealthier but de-industrializing countries like the United States and the United Kingdom. The impact of these economic changes on workers was a hotly debated factor in the modern politics of developed nations.⁸²

Brynjolfsson's "The Turing Trap" Essay (2022)

Erik Brynjolfsson, a prominent researcher on the social impacts of innovation, published an essay addressing concerns about the future of an AI-driven society. He argued that generative AI could take two paths: one that automates human labor out of existence, or one that augments and enhances it. Societies that prioritize automation might see short-term economic gains benefiting those in leadership, but at the cost of long-term consequences for the broader population. This concentration of wealth and power in the hands of a few would drive future innovation further toward automation, reinforcing inequality and limiting opportunities for the less powerful.⁸³

The concept of this phenomenon as a "Turing Trap" stems from Alan Turing's foundational work on artificial intelligence. Turing's work inspired the goal of creating machines that can fully replicate human intelligence, but Brynjolfsson warns that this focus can lead societies to prioritize replacing human labor entirely, rather than augmenting it. Over time, if this path is carried out without the voice of workers, the bargaining power of the replaced workers is reduced to the point that they are powerless to influence policy. According to Brynjolfsson, generative AI will boost labor productivity. However, "no economic law ensures everyone will share this growing pie."⁸⁴

If the default path of this technological revolution under current economic incentives is to replace human workers with machines, the author argues that it is critical for policymakers to alter the incentives to promote the path of augmentation, which he views as the best path for society as a whole. Brynjolfsson published a follow-up essay in 2024 affirming that AI augmentation held more productivity benefits than machines or humans working alone, especially benefitting less-experienced

workers. Approaches that encouraged human-machine collaboration, keeping humans in the job while increasing their capacity for efficient, effective work, had the potential to lead to more equally-distributed incomes.

Agarwal, Gans, and Goldfarb's "The Turing Transformation" Paper (2023)

A group of University of Toronto economists argued that AI-driven task automation can enhance job prospects and expand employment by democratizing access to certain high-skill tasks, reducing the "skill premium" tied to higher wages.⁸⁵ By making once-scarce skills more widely available, the authors argue, new opportunities will be created for lower-wage workers, leveling the economic playing field.

They illustrate this theory with an example of a once-scarce professional skill that was democratized by technology: the knowledge of local streets that enabled taxi drivers to navigate without maps. What was once the exclusive preserve of career drivers has been widely distributed by GPS systems in the smartphone era, allowing many more people to do the job. Perhaps these new drivers don't perform as well as career drivers initially, but they are capable enough to transport customers as ridesharing contractors. As app-based ride-sharing services grew the supply of new drivers, traditional taxi driver wages fell.

If the main consideration is the impact on licensed taxi drivers, this development appears to result in exactly the kind of reduction in bargaining power and de-skilling through automation that other economists have warned against. But when one widens the scope to consider the impact on workers lower down the skills and earnings ladder, the overall impact is an increase in economic opportunity across the population.⁸⁶ When technology automated a once-scarce skill, the total number of drivers earning some income from driving exploded, as did the number of people willing to pay for a ride, creating a net gain in economic activity. Though career drivers saw a reduction in earnings from the new competition, many new jobs were created for those at the lower end of the skills distribution. The same transformation that was bad for some workers created new opportunities for others.⁸⁷

The paper also argues that the dichotomy between automation and augmentation is misleading. A technology that augments the labor of one kind of worker may automate away the work of another, regardless of its initial intent. As the authors put it: "One person's augmentation is another's automation."⁸⁸ Office productivity software, which can be seen as augmenting the work of some traditionally high-earning knowledge workers, also automated away a number of traditionally lower-earning administrative workers who were no longer needed to answer phones or type documents. Flipping traditional arguments about automation and inequality on their heads, the authors argue that "the winners and losers are not determined by whether the technology seems to replace or augment human tasks ... the winners and losers are determined by whether the augmentation affects lower-wage workers and automation affects those already doing well."⁸⁹

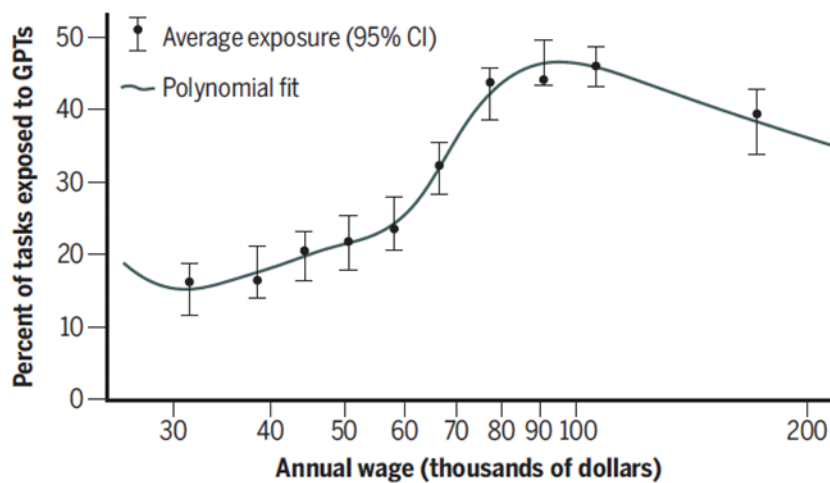
Looking Ahead

In an interview with MIT President Sally Kornbluth, OpenAI CEO Sam Altman expressed a measured yet decidedly optimistic view on the future of AI, despite growing concerns about its potential disruption. Referencing AI as "the biggest, the best, and the most important" technology revolution yet, he stated:

AI will continue to get way more capable... People are using it to create amazing things. If we could see what each of us can do 10 or 20 years in the future, it would astonish

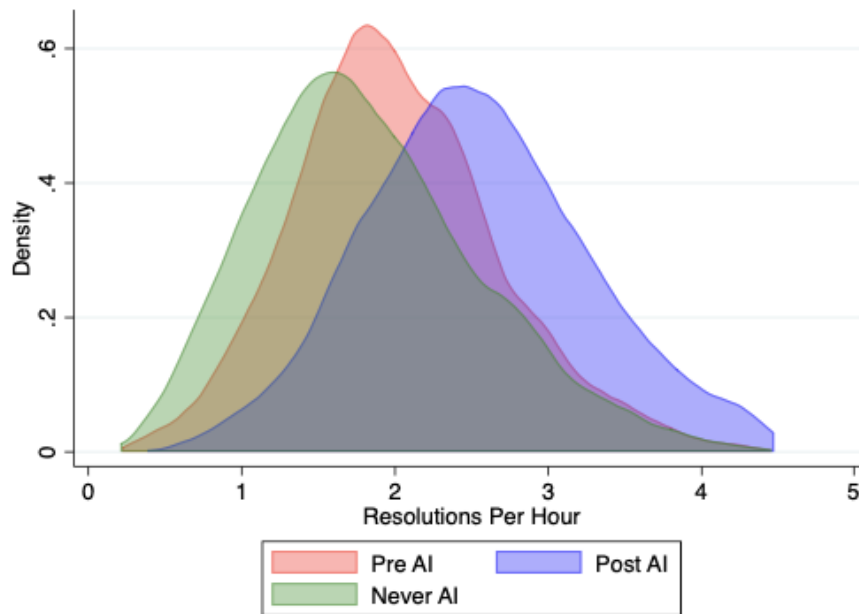
us today... Society always holds space for doomsayers. There's value to that. I'm happy that they exist. I think it makes us think harder about what we're doing. But I think the better question is, what needs to happen to navigate [AI] safety sufficiently well? AI is going to eliminate a lot of current jobs, and there will be classes of jobs that totally go away. AI is also going to change the way a lot of current jobs function, and it's going to create entirely new jobs. We'll find that humans are wired to care about other humans. As long as you believe humans want to create and be useful, we're not going to run out of things to do.⁹⁰

Altman's remarks offer a hopeful yet realistic perspective on AI's future. While acknowledging that AI will disrupt the labor market, he emphasizes that it will also create new opportunities for work and human creativity. His confidence in humanity's resilience frames AI as a tool that, with careful management, can enhance rather than diminish human contributions to society. As AI capabilities continue to advance, Altman's balanced view underscores the importance of thoughtful navigation of both its risks and rewards.

Exhibit 1 Exposure to GPTs by Income

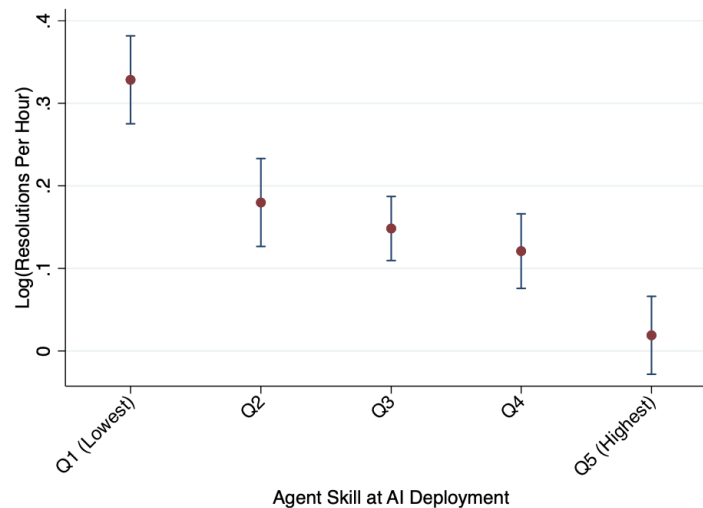
CI, confidence interval; GPT, generative pretrained transformer; LLM, large language model.

Source: Tyna Eloundou, Sam Manning, Pamela Mishkin, and Daniel Rock, “GPTs are GPTs: Labor Market Impact Potential of LLMs,” *Science* 384, 1306-1308 (2024), [10.1126/science.adj0998](https://doi.org/10.1126/science.adj0998).

Exhibit 2 Brynjolfsson, Li, and Raymond (2023): Resolutions per Hour across Three Conditions:

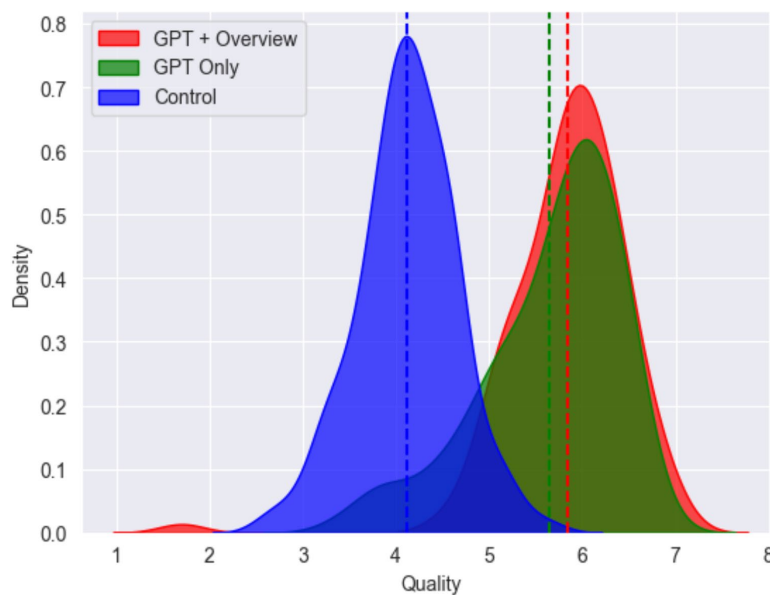
Source: Erik Brynjolfsson, Danielle Li, and Lindsey R. Raymond, "Generative AI at Work," NBER Working Paper 31161, April, 2023, rev Nov 2023, <https://www.nber.org/papers/w31161>. © 2023 by Erik Brynjolfsson, Danielle Li, and Lindsey R. Raymond. All rights reserved.

Exhibit 3 Brynjolfsson, Li, and Raymond (2023): Impact of AI on Resolutions per Hour, by Skill Quintile at Deployment

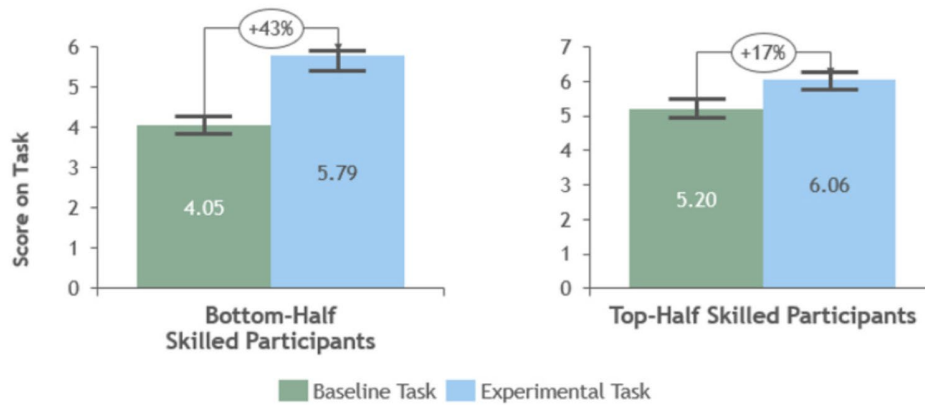


Source: Erik Brynjolfsson, Danielle Li, and Lindsey R. Raymond, “Generative AI at Work,” NBER Working Paper 31161, Apr, 2023, rev Nov 2023, <https://www.nber.org/papers/w31161>. © 2023 by Erik Brynjolfsson, Danielle Li, and Lindsey R. Raymond. All rights reserved.

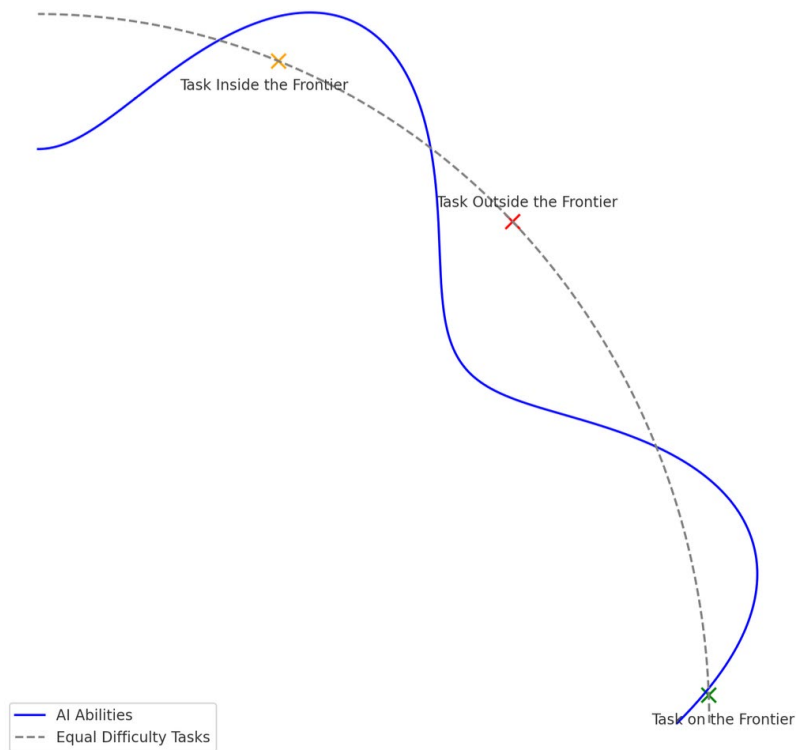
Exhibit 4 Dell’Acqua et al. (2023): Performance Distribution – Inside the Frontier



Source: Fabrizio Dell’Acqua, Edward McFowland, Ethan R. Mollick, Hila Lifshitz-Assaf, Katherine Kellogg, Lisa Kraye, François Candelon, and Karim R. Lakhani, “Navigating the Jagged Technological Frontier: Field Experimental Evidence of the Effects of AI on Knowledge Worker Productivity and Quality,” HBS Working Paper, Sep 18, 2023 (rev Sep 27, 2023), https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4573321, accessed Nov 13, 2023.

Exhibit 5 Dell'Acqua et al. (2023): Bottom-Half Skills and Top-Half Skills – Inside the Frontier

Source: Fabrizio Dell'Acqua, Edward McFowland, Ethan R. Mollick, Hila Lifshitz-Assaf, Katherine Kellogg, Lisa Kraye, François Candelon, and Karim R. Lakhani, "Navigating the Jagged Technological Frontier: Field Experimental Evidence of the Effects of AI on Knowledge Worker Productivity and Quality," HBS Working Paper, Sep 18, 2023 (rev Sep 27, 2023), https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4573321, accessed Nov 13, 2023.

Exhibit 6 Dell’Acqua et al. (2023): Illustration of the “Jagged Frontier”

Source: Fabrizio Dell’Acqua, Edward McFowland, Ethan R. Mollick, Hila Lifshitz-Assaf, Katherine Kellogg, Lisa Kraymer, François Candelon, and Karim R. Lakhani, “Navigating the Jagged Technological Frontier: Field Experimental Evidence of the Effects of AI on Knowledge Worker Productivity and Quality,” HBS Working Paper, Sep 18, 2023 (rev Sep 27, 2023), https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4573321, accessed Nov 13, 2023.

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